

Electric Vehicle Routing Problem with Overnight Depot Charging and En-Route Charging

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Appendix A. Summary of Symbols

Table A.1: Sets, parameters, and decision variables of the proposed MILP

Set	Description
C	set of customer nodes
F	set of charging station nodes
N	set of all nodes (depot, customers and charging stations)
A	set of arcs
$\mathcal{P}$	set of paths between any two nodes
$\mathcal{P}^{\text{DC}}$	set of depot-to-customer paths
$\mathcal{P}^{\text{CC}}$	set of customer-to-customer paths
$\mathcal{P}^{\text{CD}}$	set of customer-to-depot paths
$\mathcal{P}^{\text{D}}$	set of direct paths
$\mathcal{P}^{\text{CH}}$	set of charging paths
$\mathcal{L}_p$	index set $\{1, \dots, n(p)\}$ of public-CS visits along path $p \in \mathcal{P}^{\text{CH}}$
O	set of all overnight charging operations on the path set $\mathcal{P}^{\text{DC}}$
$\mathcal{A}^{\text{OC}}$	retained sequencing arcs between the dummy source/sink and real depot charging operations
Parameter	Description
q_i	demand of customer i
s_i	service time of customer i
$[\alpha_i, \beta_i]$	time window of customer i
T	end of the planning horizon
T_B	earliest allowable departure time for daytime routing
$[E_{\min}, E_{\max}]$	prescribed operating SOC range for EV
Q	load capacity of EV
ρ_f	energy charging rate at public CSs (energy per unit of model time)
ρ_s	energy charging rate at depot chargers (energy per unit of model time)
$\mathcal{B}$	number of available depot chargers
d_{ij}	travel distance from node i to node j , used by the large-scale ReplacePath rule
t_{ij}, c_{ij}	travel time and energy consumption between (i, j)
δ_p^-, δ_p^+	origin and destination of path $p \in \mathcal{P}$
$n(p)$	number of CSs visited along path $p \in \mathcal{P}$
e_p	energy consumption incurred when traversing the path $p \in \mathcal{P}$
t_p	travel time (excluding any charging time) when traversing the path $p \in \mathcal{P}$
$cs(p, \ell)$	ℓ^{th} CS visited along path $p \in \mathcal{P}^{\text{CH}}$
(o^-, o^+)	source and sink dummy operations of the depot-charging sequencing network
Variable	Description
x_p	1 if path $p \in \mathcal{P}$ is selected; 0 otherwise
τ_p	departure time from the origin node of path p
r_p	departure SOC from the origin node of path p
$\underline{r}_{p\ell}, \bar{r}_{p\ell}$	arriving SOC and departing SOC of the EV at ℓ^{th} CS on path $p \in \mathcal{P}^{\text{CH}}$
Δ_p	total charging time incurred along path $p \in \mathcal{P}^{\text{CH}}$
$\underline{t}_p, \bar{t}_p$	start time and end time of the overnight charging operation for path $p \in \mathcal{P}^{\text{DC}}$
$f_{o'o}$	1 if operation o' immediately precedes operation o on the same charger; 0 otherwise
u_i	remaining vehicle load after serving customer i

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Appendix B. Detailed results for the EVRP-OC instances

Table B.1: Detailed results on small-scale instances under $\mathcal{B} = 1$

Instance	MILP			Matheuristic					
	Obj	Gap	Time (s)	Best 10	Avg 10	Time (s)	#CS	Dist.	#NV
C101C5	907.19	0.00%	1.70	907.19	908.30	0.13	3	264.44	2
C103C5	394.96	0.00%	0.43	394.96	394.96	0.17	1	172.97	2
C206C5	572.76	0.00%	10.13	572.76	572.76	0.21	2	244.37	2
C208C5	425.39	0.00%	1.52	425.39	425.39	0.16	2	205.67	2
R104C5	182.93	0.00%	0.99	182.93	182.93	0.16	2	149.26	2
R105C5	226.51	0.00%	0.79	226.51	226.51	0.20	3	177.49	2
R202C5	162.17	0.00%	1.99	162.17	162.17	0.23	2	128.78	1
R203C5	232.07	0.00%	1.17	232.07	232.07	0.30	3	195.63	2
RC105C5	294.28	0.00%	2.41	294.28	294.28	0.27	3	243.09	3
RC108C5	335.98	0.00%	1.43	335.98	335.98	0.15	3	269.74	2
RC204C5	193.06	0.00%	1.09	193.06	193.06	0.20	1	187.69	3
RC208C5	203.17	0.00%	2.37	203.17	203.17	0.18	2	167.98	1
C101C10	INF	—	10.77	—	—	—	—	—	—
C104C10	948.41	30.00%	7200.13	948.41	948.41	0.49	4	282.67	2
C202C10	672.89	0.00%	3543.25	672.89	672.89	0.89	2	275.39	3
C205C10	955.62	0.00%	550.01	955.62	1083.25	0.60	4	285.98	2
R102C10	INF	—	9.70	—	—	—	—	—	—
R103C10	267.84	0.00%	183.58	267.84	267.84	0.85	2	210.79	3
R201C10	261.51	0.00%	12.57	261.51	261.51	1.29	2	228.36	3
R203C10	287.28	0.00%	26.54	287.28	287.28	0.78	3	232.68	2
RC102C10	INF	—	0.10	—	—	—	—	—	—
RC108C10	INF	—	1173.91	—	—	—	—	—	—
RC201C10	381.19	0.00%	13.96	381.19	381.19	1.45	4	312.57	3
RC205C10	392.46	0.00%	2.67	392.46	392.46	1.32	3	325.98	2
C103C15	NFS	—	7200.09	1668.12	1668.12	1.04	8	433.54	3
C106C15	NFS	—	7200.09	NFS	NFS	0.71	NFS	NFS	NFS
C202C15	NFS	—	7200.06	NFS	NFS	1.52	NFS	NFS	NFS
C208C15	1100.56	41.19%	7200.08	1100.56	1100.56	1.82	4	341.54	3
R102C15	INF	—	1892.36	—	—	—	—	—	—
R105C15	NFS	—	7200.33	NFS	NFS	1.83	NFS	NFS	NFS
R202C15	485.22	0.00%	4122.05	485.22	485.22	4.86	6	378.14	4
R209C15	374.53	0.00%	188.63	374.53	374.53	2.98	3	307.68	3
RC103C15	NFS	—	7200.05	NFS	NFS	1.31	NFS	NFS	NFS
RC108C15	NFS	—	7200.06	NFS	NFS	1.13	NFS	NFS	NFS
RC202C15	472.92	0.00%	95.90	472.92	473.47	4.31	5	405.28	4
RC204C15	380.05	8.45%	7200.06	380.05	380.21	1.37	4	312.00	2

INF: infeasibility proved by the MILP; NFS: no feasible solution found within the time limit; —: not available/not applicable. Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.2: Detailed results on small-scale instances under $\mathcal{B} = 2$

Instance	MILP			Matheuristic					
	Obj	Gap	Time (s)	Best 10	Avg 10	Time (s)	#CS	Dist.	#NV
C101C5	491.57	0.00%	0.97	491.57	491.57	0.38	2	253.21	4
C103C5	165.67	0.00%	0.95	165.67	165.67	0.25	0	165.67	3
C206C5	331.28	0.00%	0.69	331.28	331.28	0.48	1	268.42	4
C208C5	376.77	0.00%	1.51	376.77	376.77	0.24	2	205.00	2
R104C5	146.17	0.00%	0.12	146.17	146.17	0.18	1	136.69	2
R105C5	177.54	0.00%	0.11	177.54	177.54	0.35	2	156.08	2
R202C5	162.17	0.00%	1.78	162.17	162.17	0.24	2	128.78	1
R203C5	232.07	0.00%	1.36	232.07	232.07	0.32	3	195.63	2
RC105C5	259.57	0.00%	1.08	259.57	261.03	0.45	3	238.18	3
RC108C5	292.32	0.00%	0.63	292.32	292.32	0.29	3	253.93	2
RC204C5	193.06	0.00%	1.41	193.06	193.06	0.25	1	187.69	3
RC208C5	203.17	0.00%	1.18	203.17	203.17	0.23	2	167.98	1
C101C10	1198.22	0.00%	47.14	1198.22	1207.97	1.27	5	405.15	4
C104C10	684.89	4.36%	7200.11	684.89	684.89	1.05	3	273.93	2
C202C10	317.01	0.00%	14.99	317.01	317.01	1.91	1	291.19	4
C205C10	480.83	0.00%	22.21	480.83	480.83	1.10	2	228.28	2
R102C10	304.93	0.00%	5.67	304.93	304.93	1.79	3	253.35	3
R103C10	226.87	0.00%	21.10	226.87	226.87	1.48	1	207.87	4
R201C10	252.10	0.00%	5.61	252.10	255.41	2.10	2	224.54	3
R203C10	287.28	0.00%	21.67	287.28	287.28	1.07	3	232.68	2
RC102C10	INF	–	13.50	–	–	–	–	–	–
RC108C10	451.78	0.00%	30.00	451.78	451.78	2.16	4	387.65	4
RC201C10	347.37	0.00%	3.57	347.37	349.92	2.83	3	310.06	3
RC205C10	383.55	0.00%	2.24	383.55	383.55	1.94	3	341.38	3
C103C15	1152.58	62.62%	7200.17	1037.78	1037.78	3.87	5	368.60	3
C106C15	701.43	19.48%	7200.08	701.43	701.43	3.97	3	281.82	3
C202C15	1029.65	42.49%	7200.07	1029.65	1029.65	5.49	5	431.17	4
C208C15	673.78	0.00%	372.63	673.78	673.78	4.08	3	351.26	4
R102C15	586.13	0.00%	5341.15	586.13	595.76	5.89	8	441.46	5
R105C15	468.44	0.00%	910.30	468.44	468.44	5.68	5	373.17	5
R202C15	455.51	0.00%	823.25	455.51	455.51	6.39	6	378.14	4
R209C15	374.53	0.00%	71.27	374.53	374.53	5.55	3	307.68	3
RC103C15	473.22	0.00%	1191.10	473.22	473.22	5.64	4	397.75	5
RC108C15	461.49	0.00%	3288.46	461.49	461.67	4.80	4	379.51	3
RC202C15	462.88	0.00%	28.54	462.88	462.88	6.75	4	418.67	4
RC204C15	371.05	4.47%	7200.21	371.05	371.05	1.69	4	310.58	2

INF: infeasibility proved by the MILP; NFS: no feasible solution found within the time limit; –: not available/not applicable. Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.3: Detailed results on small-scale instances under $\mathcal{B} = 3$

Instance	MILP			Matheuristic					
	Obj	Gap	Time (s)	Best 10	Avg 10	Time (s)	#CS	Dist.	#NV
C101C5	348.97	0.00%	0.67	348.97	348.97	0.53	1	250.04	4
C103C5	165.67	0.00%	0.84	165.67	165.67	0.25	0	165.67	3
C206C5	331.28	0.00%	1.25	331.28	331.28	0.57	1	268.42	4
C208C5	376.77	0.00%	3.21	376.77	376.77	0.25	2	205.00	2
R104C5	146.17	0.00%	0.17	146.17	146.17	0.24	1	136.69	2
R105C5	177.54	0.00%	0.09	177.54	177.54	0.42	2	156.08	2
R202C5	162.17	0.00%	1.74	162.17	162.17	0.23	2	128.78	1
R203C5	232.07	0.00%	1.32	232.07	232.07	0.35	3	195.63	2
RC105C5	246.93	0.00%	0.36	246.93	246.93	0.44	1	238.05	3
RC108C5	292.32	0.00%	0.54	292.32	292.32	0.30	3	253.93	2
RC204C5	193.06	0.00%	0.94	193.06	193.06	0.28	1	187.69	3
RC208C5	203.17	0.00%	1.58	203.17	203.17	0.26	2	167.98	1
C101C10	888.52	0.00%	158.24	888.52	890.23	2.18	4	412.64	5
C104C10	575.70	0.00%	323.25	575.70	575.70	1.67	3	309.86	3
C202C10	317.01	0.00%	11.89	317.01	317.01	2.90	1	291.19	4
C205C10	374.46	0.00%	4.86	374.46	374.46	1.75	2	260.52	3
R102C10	289.56	0.00%	1.81	289.56	289.56	2.25	2	262.92	4
R103C10	211.14	0.00%	9.73	211.14	211.14	1.77	1	207.87	4
R201C10	250.33	0.00%	4.91	250.33	250.33	2.65	2	224.54	3
R203C10	287.28	0.00%	23.79	287.28	287.28	1.27	3	232.68	2
RC102C10	498.14	0.00%	3.68	498.14	498.14	3.45	4	429.66	4
RC108C10	393.23	0.00%	4.30	393.23	393.23	2.69	3	345.93	3
RC201C10	340.01	0.00%	2.71	340.01	340.01	3.03	3	310.06	3
RC205C10	383.55	0.00%	2.18	383.55	383.55	2.49	3	341.38	3
C103C15	738.31	34.51%	7200.06	738.31	738.31	6.32	5	369.54	4
C106C15	440.31	0.00%	21.94	440.31	455.18	6.38	1	306.49	5
C202C15	764.99	7.45%	7200.07	764.99	764.99	9.76	3	491.04	6
C208C15	611.95	0.00%	127.07	611.95	617.90	6.59	3	372.14	4
R102C15	510.09	0.00%	289.13	510.09	511.36	10.59	6	412.78	5
R105C15	398.58	0.00%	37.96	398.58	399.07	9.15	4	336.94	4
R202C15	444.59	0.00%	404.75	444.59	445.25	7.90	6	378.14	4
R209C15	374.53	0.00%	90.60	374.53	374.53	6.32	3	307.68	3
RC103C15	449.05	0.00%	230.99	449.05	450.82	7.87	4	397.75	5
RC108C15	423.68	0.00%	362.90	423.68	423.68	5.83	4	370.25	3
RC202C15	462.88	0.00%	27.84	462.88	463.11	8.18	4	418.67	4
RC204C15	371.05	4.45%	7200.22	371.05	371.05	2.69	4	310.58	2

Dist.: total travel distance; #CS: number of public-CS visits; #NV: number of vehicles.

Table B.4: Detailed Results for C1- and C2-Type Large-Scale Instances with Different Numbers of Depot Chargers

Instance	$\mathcal{B}$	Obj (Best 10)	Obj (Avg 10)	Time (min)	#NV	#CS	Dep. SOC	Travel Time	Charging Time
C101	12	1589.68	1589.68	3.61	12	9	95.47%	1067.11	522.57
	10	2001.19	2036.58	4.08	13	10	79.90%	1095.08	906.11
	8	2580.26	2653.76	4.13	14	13	62.13%	1123.02	1457.24
C102	12	1534.05	1534.05	3.62	12	8	92.34%	1031.33	502.73
	10	1787.50	1850.94	4.04	14	8	78.03%	1079.45	708.05
	8	2530.10	2612.76	4.29	13	13	65.52%	1100.47	1429.63
C103	12	1526.79	1527.98	3.66	12	8	90.79%	1018.23	508.56
	10	1667.43	1701.41	3.99	12	7	87.16%	1023.47	643.96
	8	2324.86	2439.54	4.31	13	11	69.34%	1084.14	1240.72
C104	12	1345.35	1346.52	3.41	11	7	89.60%	912.96	432.40
	10	1350.65	1366.20	3.81	10	8	95.53%	895.53	455.12
	8	2015.56	2213.91	4.97	15	8	67.62%	1083.33	932.23
C105	12	1555.15	1555.15	3.67	12	8	93.58%	1045.27	509.88
	10	1928.96	1960.63	4.12	13	8	79.06%	1061.46	867.50
	8	2465.43	2532.75	4.36	14	12	63.58%	1109.33	1356.10
C106	12	1553.70	1553.70	3.74	12	8	93.43%	1043.81	509.88
	10	1859.02	1891.20	4.16	13	8	80.66%	1068.73	790.29
	8	2460.63	2602.11	4.51	13	11	65.33%	1083.11	1377.52
C107	12	1554.19	1554.19	3.60	12	8	92.99%	1040.74	513.44
	10	1876.15	1919.08	3.82	13	8	79.62%	1064.32	811.84
	8	2394.94	2432.50	4.43	13	11	65.30%	1067.95	1326.99
C108	12	1547.99	1547.99	3.69	12	8	92.65%	1036.80	511.19
	10	1781.74	1867.97	4.17	13	10	80.97%	1053.57	728.17
	8	2478.88	2557.81	4.42	13	13	66.56%	1097.13	1381.75
C109	12	1421.27	1425.21	3.80	11	7	92.65%	950.90	470.38
	10	1780.66	1822.70	4.23	11	9	90.25%	1016.51	764.15
	8	2305.99	2399.54	4.52	13	10	67.19%	1062.77	1243.22
C201	4	958.18	966.21	2.84	7	3	83.60%	773.38	184.80
	3	1271.03	1287.08	3.49	7	5	65.04%	761.91	509.11
	2	1740.73	1768.99	3.94	8	8	42.48%	810.20	930.54
C202	4	945.78	949.27	2.82	7	3	85.37%	779.79	165.99
	3	1243.67	1273.84	3.53	7	4	68.05%	770.91	472.77
	2	1580.20	1596.57	3.70	6	6	59.21%	773.91	806.28
C203	4	929.92	936.94	2.84	7	2	86.13%	779.02	150.90
	3	1147.34	1161.68	3.02	7	4	72.34%	765.95	381.39
	2	1446.54	1454.50	3.46	6	6	61.42%	743.88	702.66
C204	4	867.21	880.04	2.84	7	3	87.60%	767.76	99.45
	3	1161.69	1180.68	3.14	7	3	83.18%	831.82	329.87
	2	1431.70	1442.80	3.75	7	5	58.04%	769.17	662.53
C205	4	952.05	956.61	2.96	6	3	86.20%	713.36	238.70
	3	1232.27	1239.70	3.46	7	4	63.68%	739.97	492.30
	2	1502.31	1516.92	3.77	6	6	54.27%	723.59	778.72
C206	4	947.17	952.00	2.93	6	3	89.65%	728.54	218.62
	3	1190.83	1207.16	3.28	6	4	75.42%	732.70	458.12
	2	1483.33	1493.04	3.69	6	5	53.56%	714.12	769.21
C207	4	923.35	944.03	2.88	7	3	87.12%	780.11	143.24
	3	1205.78	1229.98	3.52	7	4	67.29%	752.25	453.52
	2	1510.30	1515.83	3.79	7	5	48.05%	734.52	775.78
C208	4	902.97	918.52	2.91	6	3	87.52%	704.51	198.46
	3	1150.81	1175.15	3.16	7	3	71.11%	757.45	393.36
	2	1449.61	1470.47	3.42	6	4	56.12%	716.40	733.21

Table B.5: Detailed Results for R1- and R2-Type Large-Scale Instances with Different Numbers of Depot Chargers

Instance	$\mathcal{B}$	Obj (Best 10)	Obj (Avg 10)	Time (min)	#NV	#CS	Dep. SOC	Travel Time	Charging Time
R101	15	2062.80	2087.37	7.67	21	23	77.02%	1719.76	343.05
	12	NFS	NFS	6.86	–	–	–	–	–
	9	NFS	NFS	6.45	–	–	–	–	–
R102	15	1776.25	1786.59	6.34	18	20	86.93%	1515.53	260.72
	12	1877.73	1940.37	6.60	19	23	68.97%	1532.82	344.91
	9	NFS	NFS	5.89	–	–	–	–	–
R103	15	1424.88	1435.87	5.51	15	15	94.08%	1247.17	177.70
	12	1587.56	1605.31	5.48	16	16	81.75%	1334.74	252.82
	9	1781.33	1781.33	5.16	16	22	62.35%	1404.64	376.69
R104	15	1196.46	1205.25	4.49	12	12	97.32%	1068.52	127.94
	12	1205.35	1207.31	4.46	12	12	99.62%	1080.40	124.95
	9	1349.61	1367.57	5.57	13	13	75.18%	1134.44	215.17
R105	15	1611.26	1620.13	5.93	17	19	91.47%	1402.09	209.17
	12	1709.35	1719.73	8.37	18	18	75.24%	1427.91	281.44
	9	1844.32	1889.96	7.17	17	22	60.17%	1452.33	391.99
R106	15	1481.15	1487.08	5.26	15	16	95.94%	1292.94	188.21
	12	1585.23	1601.81	6.05	16	16	83.01%	1340.76	244.47
	9	1761.09	1768.93	8.15	17	20	61.91%	1403.60	357.49
R107	15	1281.10	1287.66	4.80	13	15	95.45%	1138.75	142.35
	12	1293.38	1303.12	4.66	12	13	97.84%	1133.48	159.90
	9	1435.18	1456.10	6.78	14	14	74.43%	1204.12	231.06
R108	15	1184.82	1192.51	4.58	12	13	97.07%	1044.58	140.24
	12	1184.82	1188.70	4.51	12	13	97.07%	1044.58	140.24
	9	1326.53	1344.89	5.92	12	14	81.06%	1101.64	224.89
R109	15	1321.65	1345.20	5.28	14	14	98.34%	1187.94	133.71
	12	1439.74	1443.44	6.31	16	14	81.80%	1254.87	184.88
	9	1514.36	1549.06	8.44	15	17	68.36%	1247.77	266.59
R110	15	1223.23	1245.82	4.73	13	12	97.82%	1108.50	114.73
	12	1259.67	1282.19	4.90	12	15	96.80%	1110.69	148.98
	9	1388.32	1399.65	10.36	14	13	72.59%	1169.16	219.16
R111	15	1249.07	1260.19	4.80	13	14	97.23%	1117.56	131.51
	12	1282.89	1294.57	5.02	13	15	94.49%	1133.34	149.55
	9	1400.51	1419.57	8.63	14	14	73.81%	1173.47	227.03
R112	15	1154.92	1165.44	4.61	12	14	98.61%	1035.18	119.74
	12	1154.92	1162.49	4.55	12	14	98.61%	1035.18	119.74
	9	1308.94	1322.73	7.52	12	14	80.41%	1095.59	213.35
R201	3	1107.61	1107.98	2.73	8	0	73.70%	1107.61	0.00
	2	1164.94	1168.44	4.39	7	4	65.81%	1123.62	41.32
	1	1238.11	1243.33	4.23	7	7	41.25%	1142.15	95.96
R202	3	993.99	998.95	2.46	6	0	70.92%	993.99	0.00
	2	1017.17	1020.55	3.11	7	1	59.14%	1013.66	3.52
	1	1068.85	1083.50	4.10	6	4	36.44%	1005.84	63.01
R203	3	875.29	880.27	2.59	6	1	77.29%	874.74	0.54
	2	897.73	905.48	3.31	6	3	71.47%	885.04	12.69
	1	955.61	966.21	3.40	5	5	52.30%	891.58	64.03
R204	3	732.97	738.82	2.32	4	0	78.15%	732.97	0.00
	2	735.91	740.79	2.31	4	0	74.66%	735.91	0.00
	1	772.47	776.40	2.43	6	0	51.98%	772.47	0.00
R205	3	960.41	967.76	2.66	6	1	80.03%	959.70	0.71
	2	968.47	977.19	3.31	6	2	80.24%	967.04	1.43
	1	1043.04	1054.77	3.69	6	5	51.92%	987.80	55.24
R206	3	907.10	910.44	2.90	6	2	82.35%	905.41	1.69
	2	920.09	926.70	3.03	6	2	76.90%	907.90	12.19
	1	1004.85	1017.90	3.84	5	4	60.22%	938.13	66.71
R207	3	795.51	811.20	2.37	5	0	59.55%	795.51	0.00
	2	795.51	802.39	2.33	5	0	59.55%	795.51	0.00
	1	818.25	840.36	3.04	6	1	50.94%	818.08	0.17
R208	3	720.80	725.18	2.50	4	1	77.59%	715.38	5.42
	2	729.22	736.50	2.36	4	1	83.04%	728.60	0.62
	1	765.13	768.87	3.16	4	1	74.41%	750.87	14.27
R209	3	872.20	874.94	2.30	6	1	79.72%	871.86	0.34
	2	873.37	879.21	3.09	6	0	80.05%	873.37	0.00
	1	941.47	951.63	3.84	4	5	70.28%	882.11	59.36
R210	3	842.56	849.00	2.46	5	2	87.52%	839.74	2.82
	2	848.67	855.03	2.70	5	2	85.65%	842.59	6.09
	1	902.14	912.62	3.45	5	5	52.85%	846.18	55.96
R211	3	760.16	772.29	2.45	4	0	71.52%	760.16	0.00
	2	765.78	776.09	2.40	5	0	57.64%	765.78	0.00
	1	793.55	804.92	2.93	5	2	56.67%	789.52	4.03

NFS: no feasible solution found.

Table B.6: Detailed Results for RC1- and RC2-Type Large-Scale Instances with Different Numbers of Depot Chargers

Instance	$\mathcal{B}$	Obj (Best 10)	Obj (Avg 10)	Time (min)	#NV	#CS	Dep. SOC	Travel Time	Charging Time
RC101	13	1989.67	2028.73	10.37	18	18	78.54%	1751.95	237.72
	11	2099.93	2119.02	10.34	19	19	66.85%	1800.41	299.52
	9	2388.90	2388.90	6.36	20	25	52.42%	1961.16	427.74
RC102	13	1829.51	1835.73	6.42	17	15	82.21%	1632.40	197.12
	11	1894.27	1910.96	6.42	17	16	71.65%	1639.93	254.34
	9	2024.95	2048.53	9.12	18	20	57.89%	1696.00	328.95
RC103	13	1501.30	1513.59	5.11	14	13	91.96%	1370.42	130.88
	11	1559.00	1574.01	5.82	14	13	84.26%	1388.57	170.43
	9	1636.43	1647.08	11.82	15	14	67.36%	1407.53	228.90
RC104	13	1314.85	1321.68	4.50	12	13	97.39%	1209.24	105.61
	11	1368.96	1407.03	4.65	12	14	94.25%	1240.17	128.79
	9	1477.59	1487.43	8.53	12	13	78.75%	1278.08	199.51
RC105	13	1711.76	1749.30	6.47	16	17	85.65%	1541.13	170.62
	11	1804.32	1817.08	7.35	16	16	73.74%	1566.39	237.93
	9	1968.15	1993.70	11.87	17	17	59.01%	1646.34	321.81
RC106	13	1613.91	1626.14	5.76	15	15	90.22%	1466.47	147.44
	11	1674.54	1685.81	5.46	15	14	79.30%	1474.44	200.10
	9	1776.30	1801.49	7.67	15	18	67.61%	1509.72	266.58
RC107	13	1373.73	1391.35	4.60	12	13	98.57%	1255.00	118.72
	11	1474.48	1494.33	5.46	12	14	93.22%	1313.94	160.54
	9	1582.35	1599.66	11.43	14	15	71.68%	1366.85	215.50
RC108	13	1278.60	1291.56	4.33	12	13	99.58%	1188.75	89.86
	11	1337.52	1356.37	4.42	12	12	95.45%	1220.57	116.95
	9	1462.28	1466.86	7.14	13	11	77.40%	1280.42	181.85
RC201	3	1276.53	1277.92	3.20	10	2	57.29%	1268.25	8.28
	2	1316.52	1328.06	7.87	10	5	49.93%	1284.26	32.26
	1	1376.20	1410.94	7.64	9	8	27.92%	1272.31	103.89
RC202	3	1140.10	1140.78	2.73	8	0	52.18%	1140.10	0.00
	2	1149.83	1151.18	3.45	8	0	52.62%	1149.83	0.00
	1	1203.69	1221.91	5.50	8	6	32.91%	1155.67	48.03
RC203	3	962.16	968.20	2.55	6	1	73.38%	957.50	4.66
	2	979.27	985.34	2.72	7	2	61.55%	970.08	9.19
	1	1058.10	1066.90	3.63	6	5	43.04%	994.74	63.37
RC204	3	854.39	857.40	2.74	5	3	92.96%	836.47	17.92
	2	870.66	873.66	2.90	6	3	75.02%	846.40	24.26
	1	952.51	961.93	3.56	5	7	54.26%	869.60	82.91
RC205	3	1079.12	1084.05	3.01	7	2	78.72%	1077.14	1.98
	2	1092.19	1094.66	3.28	7	3	72.45%	1078.44	13.75
	1	1183.08	1188.62	3.87	7	7	42.49%	1104.25	78.82
RC206	3	1073.35	1078.32	2.46	7	0	66.88%	1073.35	0.00
	2	1096.16	1104.35	2.89	7	0	68.30%	1096.16	0.00
	1	1163.05	1171.99	3.71	6	6	40.32%	1093.05	70.00
RC207	3	925.30	932.23	2.47	6	0	72.66%	925.30	0.00
	2	927.06	934.33	2.81	6	0	72.80%	927.06	0.00
	1	1012.31	1023.11	3.52	5	6	46.41%	948.47	63.84
RC208	3	823.63	827.87	2.47	5	2	89.66%	811.05	12.58
	2	823.63	830.41	2.84	5	2	89.66%	811.05	12.58
	1	882.84	902.69	3.94	5	4	50.11%	811.47	71.37

Appendix C. Detailed results for the JD Logistics case study

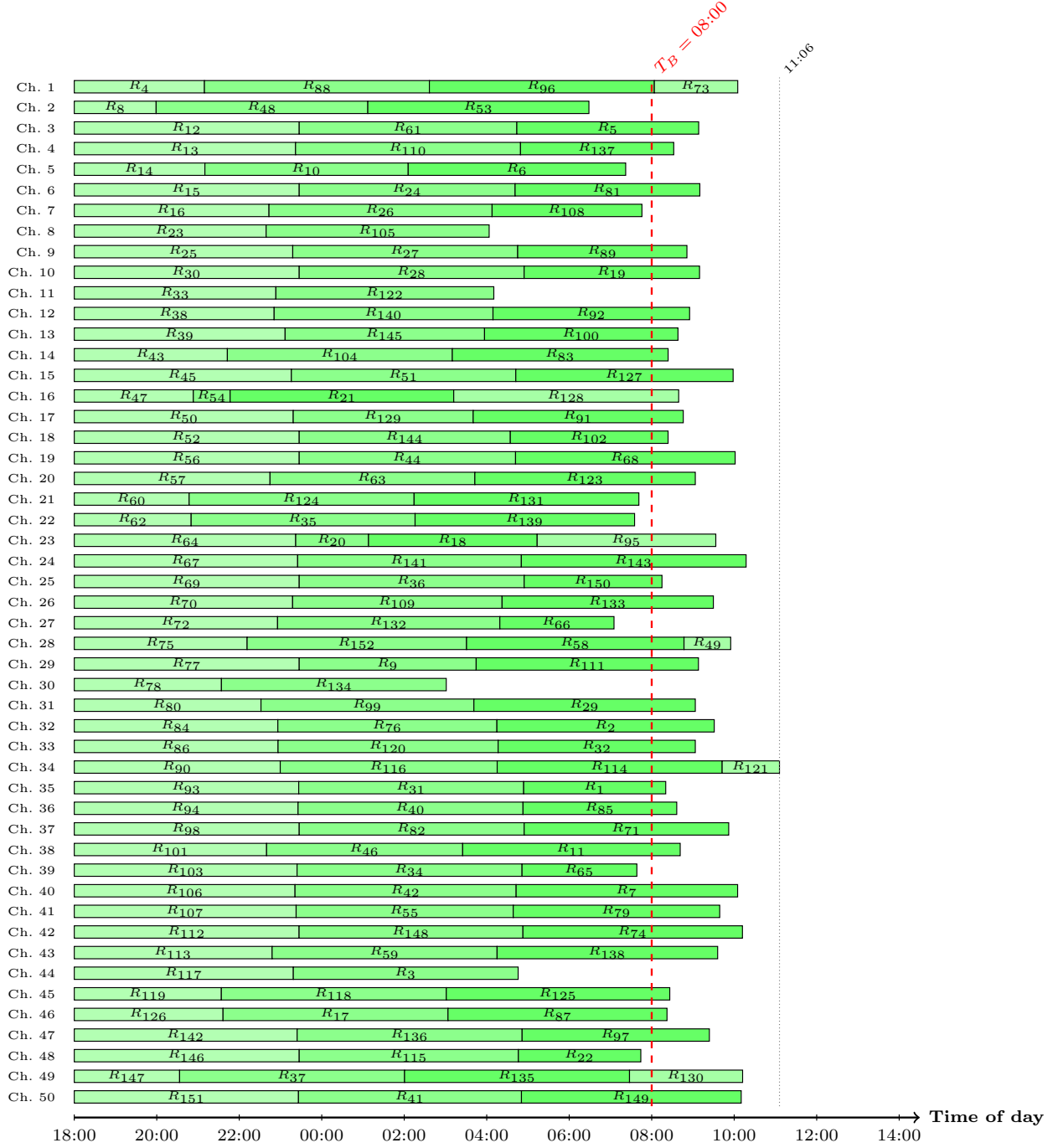


Figure C.1: Depot charging schedule for the JD Logistics case study with $B = 50$.

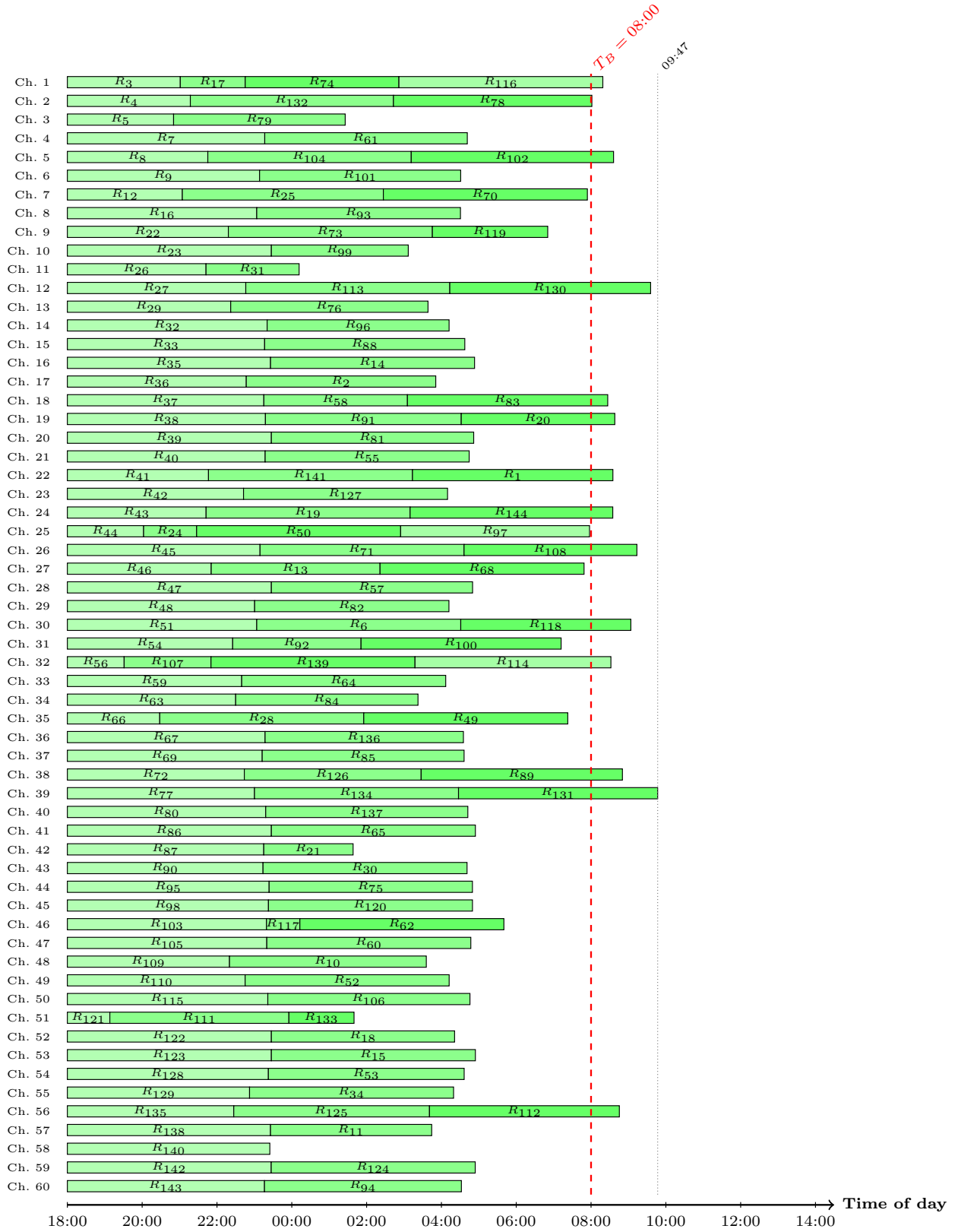


Figure C.2: Depot charging schedule for the JD Logistics case study with $B = 60$.

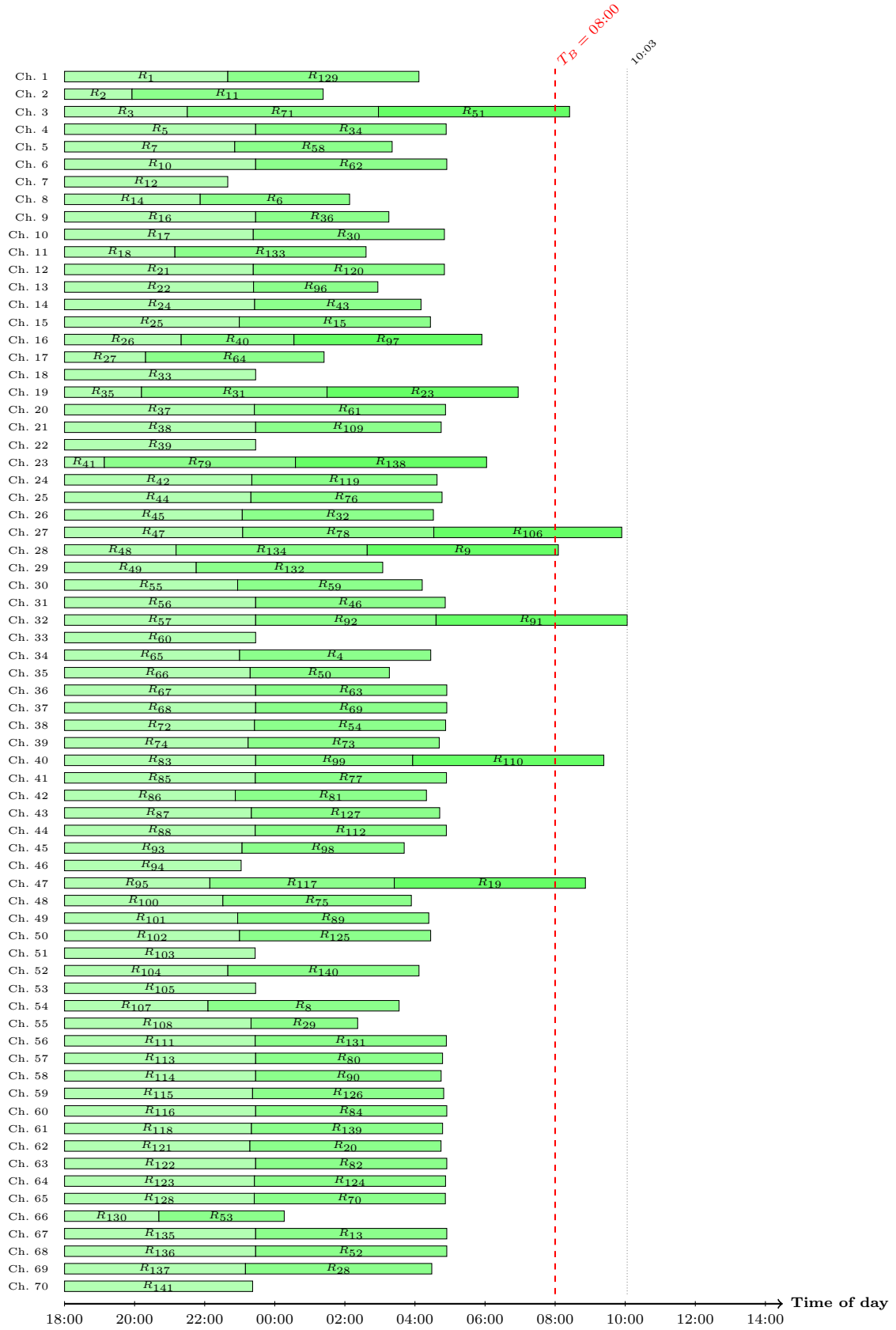


Figure C.3: Depot charging schedule for the JD Logistics case study with $\beta = 70$.

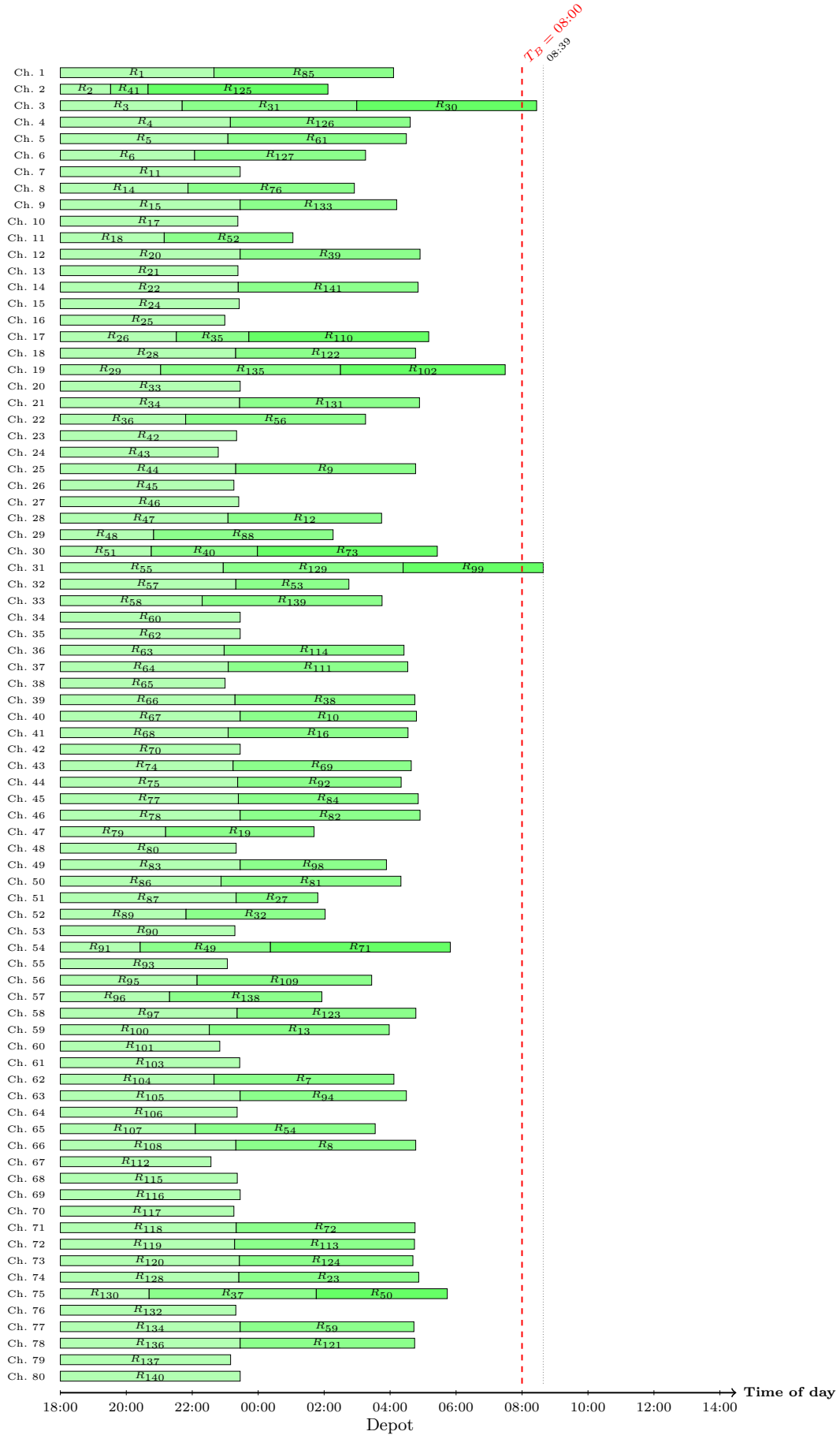


Figure C.4: Depot charging schedule for the JD Logistics case study with $\mathcal{B} = 80$.

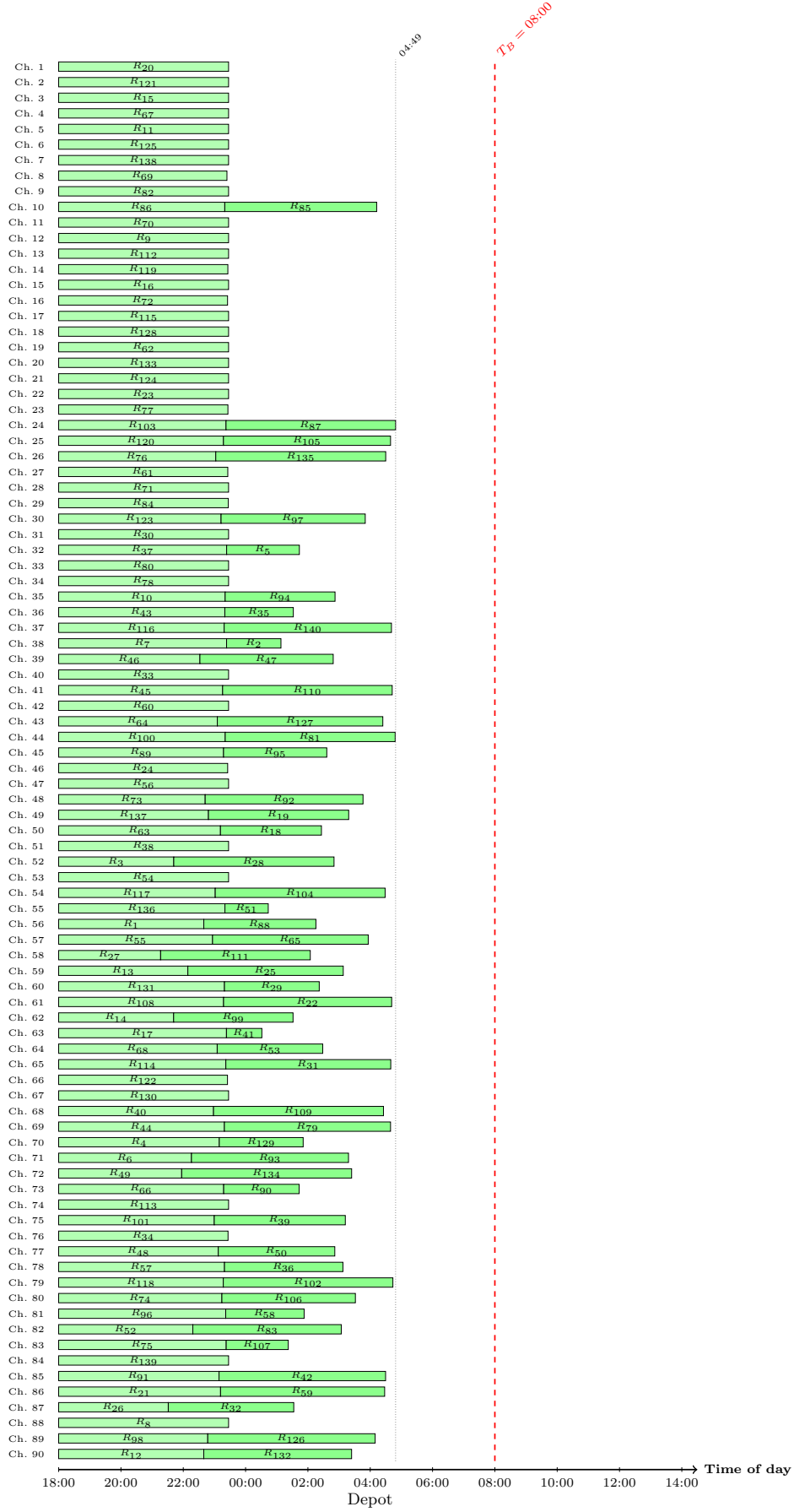


Figure C.5: Depot charging schedule for the JD Logistics case study with $B = 90$.

Appendix D. Adaptation of the matheuristic to the mE-VRSPTW

The matheuristic developed for the EVRP-OC can be adapted to solve the mE-VRSPTW of Yamín et al. (2025) while retaining its general AILS search framework. The main modifications arise from three problem-specific features of the mE-VRSPTW. First, en-route charging is not allowed, so all energy required by a route must be supplied at the depot before departure. Second, the problem is defined on a multigraph, where multiple distance–energy alternatives may exist between two consecutive vertices. Third, depot charging is performed over discrete charging periods according to a piecewise-linear charging function. Accordingly, the public-CS-specific route evaluation and continuous charging decisions of the EVRP-OC are replaced by a multigraph route evaluator and a discrete depot-charging coordinator.

The main customer-sequence search mechanisms of the AILS are retained, including randomized initialization, perturbation, adaptive penalties, and local-search neighborhoods. Since the mE-VRSPTW does not contain public CSs, the ReplacePath operator and the energy-shortage repair procedure used for en-route charging in the EVRP-OC are no longer required. The search therefore operates directly on customer sequences, using standard inter- and intra-route neighborhood moves to modify the customer assignment and visiting order.

For a fixed customer sequence, however, a route does not correspond to a unique path because each pair of consecutive vertices may be connected by multiple multigraph arcs. We therefore generate a set of nondominated realizations, hereafter referred to as *route options*. A forward labeling procedure propagates travel distance, travel time, energy consumption, and time-window feasibility along the customer sequence while considering the available arc alternatives. Dominated labels and route options are discarded. Each retained option o of route r is characterized by its total routing distance d_{ro} , energy requirement e_{ro} , latest feasible departure time $T_{ro}^{\max}$, and the corresponding overnight charging duration.

Because all energy must be supplied before departure, the charging duration of a route option is obtained from the inverse piecewise-linear charging function of the mE-VRSPTW. Following its discrete-time charging representation, the resulting duration is converted into the required number of complete charging periods. Hence, selecting a route option simultaneously determines its routing cost and the amount of scarce depot-charging capacity required before departure.

During the AILS, a route option is evaluated using a search criterion that accounts for both routing distance and overnight charging demand. In particular, the charging-duration component is weighted by an adaptive parameter λ . When the current route set is difficult to accommodate within the available depot charging capacity, λ is increased, placing greater emphasis on route options with lower energy requirements and shorter charging durations. When the depot charging schedule is readily feasible, λ is reduced, allowing the search to place greater emphasis on routing distance. The original mE-VRSPTW objective remains the minimization of total routing distance; the charging-related term is used only to guide the heuristic search. This mechanism plays a role analogous to the adaptive departure-SOC control used for the EVRP-OC.

For a selected route option, the corresponding overnight charging operation is represented by its charging duration and latest feasible completion time. The depot subproblem is therefore a nonpreemptive scheduling problem on B identical chargers, where each EV must be assigned to exactly one charger and its charging operation must be completed before the departure deadline of the selected route option. To evaluate candidate solutions efficiently during the AILS, we first construct a greedy earliest-deadline schedule. If this procedure successfully schedules all charging operations, the resulting schedule provides an explicit certificate of depot-charging feasibility. Otherwise, a bounded depth-first search is used to explore alternative charger assignments. Reaching the prescribed search limit is treated as an unresolved case rather than as proof of infeasibility; a solution is classified as depot-feasible only when a complete feasible charging schedule has been explicitly constructed.

The route-pooling idea of the original matheuristic is also retained, but the continuous SP model developed for the EVRP-OC is not transferred literally. Instead, the coordination step jointly selects multigraph route options for the customer routes and verifies their compatibility with the discrete depot-charging schedule. For small combinations of route options, the alternatives can be examined directly. For larger combinations, the implementation explores promising option selections within prescribed computational limits while retaining only solutions for which a feasible depot schedule is explicitly constructed.

Finally, when a depot-feasible solution is about to improve the global incumbent, an additional route-option refinement is performed. Starting from the validated solution, alternative combinations of nondominated route options are explored in increasing order of routing distance. A neighboring state is obtained by replacing the option of one route with another nondominated alternative, allowing coordinated changes across several routes. The incumbent objective provides an upper bound for pruning option combinations that cannot improve the current best solution. The refinement is strictly bounded by the number of explored option states, charger-assignment search effort, and computing time. If a limit is reached, the previously validated incumbent is retained.

Overall, the adaptation preserves the route-generation and intensification philosophy of the EVRP-OC matheuristic while replacing the continuous departure-SOC and public-charging decisions by multigraph route-option selection and discrete overnight charging

coordination. This enables the same general framework to exploit the distance–energy trade-off specific to the mE-VRSPTW while explicitly accounting for the limited charging capacity at the depot.

References

Yamín, D., Desaulniers, G., & Mendoza, J. E. (2025). The electric vehicle routing and overnight charging scheduling problem on a multigraph. *INFORMS Journal on Computing*, 37, 808–830.